Haozhe Lei

Ph.D. Candidate, Electrical and Computer Engineering | NYU WIRELESS

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RESEARCH PROFILE

RF sensing and probabilistic spatial intelligence for autonomous systems.

- **Core problem and inference:** I formulate wireless localization as spatial-belief estimation: one multipath snapshot can support several transmitter positions and headings when antennas are directional or maps and digital twins are mismatched. I learn candidate-wise likelihoods, form calibrated spatial posteriors, and fuse observations.
- **Real-to-sim calibration and task-driven RF sensing:** I align ray-tracing wireless digital twins to measurements and learn map- and pose-conditioned neural scores for localization. I also co-developed physics-informed methods that use RF evidence for navigation under partial observability.
- **System translation:** The program spans RF localization, robotic sensing and navigation, and validation on a 10-GHz FR3 platform. Complementary directions include UE-centric multi-band adaptation and vision–language-guided object-centric RGB-D spatial memory for embodied agents.

EDUCATION

New York University Sept. 2022 – Expected May 2027
Ph.D. in Electrical and Computer Engineering
Advisor: Prof. Sundeep Rangan, Director, NYU WIRELESS *Fellowship: Ernst Weber Fellowship, 2023*

New York University Sept. 2020 – May 2022
M.S. in Computer Engineering

China Agricultural University Sept. 2015 – Jun. 2019
B.E. in Electrical Engineering and Automation

RESEARCH EXPERIENCE

NYU WIRELESS, New York University *Brooklyn, NY, USA*
Graduate Research Assistant; Advisor: Prof. Sundeep Rangan *Sept. 2022 – Present*

- **Probabilistic RF localization and measurement-aligned wireless digital twins.** Lead end-to-end development of likelihood-based and neural posterior localization under sparse measurements and digital-twin mismatch, from problem formulation and algorithm design through implementation and evaluation. Methods include uncertainty calibration, Bayesian fusion, U-Net/attention spatial scoring, and measurement-aligned Sionna RT. Lead *MAGNETAR* from probabilistic formulation through RF–robot integration and experimental evaluation, combining multipath spatial posteriors with *MobiFR3* measurements for transmitter-pose inference. *Outputs: IEEE GLOBECOM 2026; Asilomar 2026 (two papers); IEEE TVT manuscript under review; IEEE ICRA 2027 (two papers under review); IEEE TWC manuscript revised and resubmitted.*
- **Wireless-aware robotic navigation: methods and experimental systems.** Develop *MobiFR3*, a heterogeneous 10-GHz FR3 robotic sensing platform. Coordinate system architecture, RF–robot synchronization, and localization experiments with Pi-Radio, TurtleBot4/ROS 2, odometry, LiDAR, and RGB. The latest Jackal UGV configuration adds an FR3 radio payload, 3D LiDAR, and RGB-D sensing, alongside recorded SLAM and path-routing demonstrations. Mentor students in RF hardware operation and data collection. Earlier, co-developed *PIRL* and program-guided reinforcement learning for zero-shot wireless indoor navigation in simulation. *Publications: IEEE ICRA 2024; IEEE OJ-COMS 2025; and RLC IBRL Workshop 2025 (Spotlight).*
- **UE-centric wireless digital twins for closed-loop multi-band adaptation.** Lead the research direction, problem formulation, digital-twin design, and learning methods for *MCMB-HDT*, spanning georeferenced propagation, multi-cell FR1/FR3 networks, handset arrays, and mobility. Develop Transformer-based sparse-history rate prediction and recurrent PPO for measurement-constrained array activation; mentor a junior researcher in model development, Sionna implementation, and evaluation. *Outputs: IEEE ICC Workshops 2026; MCMB-HDT manuscript (venue to be determined).*

PUBLICATIONS AND MANUSCRIPTS

Haozhe Lei is shown in bold; * denotes equal contribution; † denotes corresponding author.

Peer-Reviewed Publications

Wireless Sensing, Localization, and Robotic Systems

[1] **Haozhe Lei**†, Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. "LOCUS-DT: Localization via Observation-Conditioned Uncertainty Scoring with Digital Twins". *IEEE Global Communications Conference (GLOBECOM)*. 2026. **[paper]**.

[2] **Haozhe Lei**†, Hao Guo, Tommy Svensson, and Sundeep Rangan. "Beyond Point Estimates: Likelihood-Based Full-Posterior Wireless Localization". *Asilomar Conference on Signals, Systems, and Computers*. 2026. **[paper]**.

[3] Mingsheng Yin*, Tao Li*, **Haozhe Lei***, Yaqi Hu, Sundeep Rangan, and Quanyan Zhu. "Zero-Shot Wireless Indoor Navigation through Physics-Informed Reinforcement Learning". *IEEE International Conference on Robotics and Automation (ICRA)*. 2024. **[paper]**.

[4] Tao Li, **Haozhe Lei**, Hao Guo, Mingsheng Yin, Yaqi Hu, Quanyan Zhu, and Sundeep Rangan. "Digital Twin-Enhanced Wireless Indoor Navigation: Achieving Efficient Environment Sensing with Zero-Shot Reinforcement Learning". *IEEE Open Journal of the Communications Society* (2025). **[paper]**.

[5] Roberto Bomfin, **Haozhe Lei**, Sundeep Rangan, and Marwa Chafii. "Likelihood-Based Wireless Localization with Last-Bounce Spatial Features". *Asilomar Conference on Signals, Systems, and Computers*. 2026.

[6] Tao Li, **Haozhe Lei**, Mingsheng Yin, and Yaqi Hu. "Reinforcement Learning with Physics-Informed Symbolic Program Priors for Zero-Shot Wireless Indoor Navigation". *Inductive Biases in Reinforcement Learning Workshop at RLC*. Spotlight paper. 2025. **[paper]**.

Wireless Digital Twins and Multi-Band Systems

[7] Ruibin Chen*, **Haozhe Lei***,†, Hao Guo, Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. "Transformer-Based Rate Prediction for Multi-Band Cellular Handsets". *IEEE International Conference on Communications Workshops (ICC Workshops)*. 2026. **[paper]**.

Cyber-Physical Systems: Control, Mobility, and Security

[8] Zhaoye Pan, **Haozhe Lei**, Fan Zuo, Zilin Bian, and Tao Li. "Distributed Online Convex Optimization with Nonseparable Costs and Constraints". *IEEE Control Systems Letters* 10 (2026), pp. 391–396. **[paper]**.

[9] Tao Li*, Zilin Bian*, **Haozhe Lei***, Fan Zuo*, Ya-Ting Yang*, Quanyan Zhu, Zhenning Li, Zhibin Chen, and Kaan Ozbay. "Digital Twin-Based Driver Risk-Aware Predictive Mobility Analytics for Real-Time Situational Awareness Through Cooperative Sensing". *IEEE Transactions on Intelligent Transportation Systems* (2025). **[paper]**.

[10] Ya-Ting Yang, **Haozhe Lei**, and Quanyan Zhu. "PRADA: Proactive Risk Assessment and Mitigation of Misinformed Demand Attacks on Navigational Route Recommendations". *IEEE Transactions on Information Forensics and Security* (2025). **[paper]**.

[11] Zilin Bian, Tao Li, **Haozhe Lei**, Fan Zuo, Ya-Ting Yang, Quanyan Zhu, Zhenning Li, and Kaan Ozbay. "Multi-level Traffic-Responsive Tilt Camera Surveillance through Predictive Correlated Online Learning". *Transportation Research Part C: Emerging Technologies* (2024). **[paper]**.

[12] **Haozhe Lei***,†, Yunfei Ge*, and Quanyan Zhu. "ADAPT: A Game-Theoretic and Neuro-Symbolic Framework for Automated Distributed Adaptive Penetration Testing". *IEEE Military Communications Conference (MILCOM)*. 2024. **[paper]**.

[13] Tao Li*, **Haozhe Lei***, and Quanyan Zhu. "Self-Adaptive Driving in Nonstationary Environments through Conjectural Online Lookahead Adaptation". *IEEE International Conference on Robotics and Automation (ICRA)*. 2023. **[paper]**.

Manuscripts

[M1] **Haozhe Lei**[†], Roberto Bomfin, Marwa Chafii, and Sundeep Rangan. "Posterior Inference for Indoor Localization with Ray-Tracing Wireless Digital Twins". Revised and resubmitted to IEEE Transactions on Wireless Communications. 2026.

[M2] **Haozhe Lei**[†], Hao Guo, and Sundeep Rangan. "Learning a Measurement-to-Posterior Map for Wireless Localization". IEEE Transactions on Vehicular Technology, under review. 2026.

[M3] **Haozhe Lei**[†], Ruibin Chen, Yuhan Jiang, Ali Rasteh, Aditya Dhananjay, and Sundeep Rangan. "MAGNETAR: Multipath-Guided Spatial Posteriors for Transmitter Pose Inference in the Upper Mid-Band". IEEE International Conference on Robotics and Automation (ICRA), under review. 2027. **[paper]**.

[M4] **Haozhe Lei**[†] and Sundeep Rangan. "MAPLE-RF: Efficient Probabilistic RF Source Localization in Partially Explored Environments". IEEE International Conference on Robotics and Automation (ICRA), under review. 2027. **[paper]**.

[M5] Ruibin Chen[*], **Haozhe Lei**[*,†], Marco Mezzavilla, Hitesh Poddar, Tomoki Yoshimura, and Sundeep Rangan. "MCMB-HDT: A Multi-Cell Multi-Band Handset Digital Twin for Learning-Based Closed-Loop Array Activation". Manuscript; venue to be determined. 2026.

RESEARCH MENTORING

New York University Feb. 2025 – Present

- **Ruibin Chen**, Ph.D. student *Feb. 2025 – Present*
  Mentor his research on *MCMB-HDT* and RF localization, including model development, Sionna implementation, and evaluation, within projects I lead. Joint work includes a peer-reviewed IEEE ICC Workshops 2026 paper, the MCMB-HDT manuscript (venue to be determined), and a manuscript under review at IEEE ICRA 2027.
- **Yuhan Jiang**, M.S. student *Jul. 2026 – Present*
  Mentor research on RF hardware operation, calibration, and data acquisition for the FR3 robotic localization platform; co-author of *MAGNETAR*, under review at IEEE ICRA 2027.
- **Yuheng Liu and Ashesh Kaji**, M.S. students *2026 – Present*
  Mentor their joint project on object-centric RGB-D scene memory for embodied agents, including VLM-guided object proposals, depth-grounded 3D projection, cross-view association, and queryable scene representations for retrieval and localization. Yuheng joined in Jan. 2026; Ashesh joined in Jul. 2026.

ADDITIONAL RESEARCH EXPERIENCE

NYU C2SMARTER Center *Brooklyn, NY, USA*
Graduate Researcher *Sept. 2023 – Mar. 2026*

- Developed distributed online optimization, predictive camera control, and risk-aware transportation digital twins for networked sensing.

NYU Center for Cybersecurity *Brooklyn, NY, USA*
Graduate Researcher *Sept. 2021 – Jan. 2024*

- Developed adaptive decision-making and security methods for autonomous driving, navigation, and automated penetration testing.

TEACHING EXPERIENCE

Teaching Assistant, New York University Sept.–Dec. 2024
ECE-GY 5213 Introduction to System Engineering

TECHNICAL EXPERTISE AND EXPERIMENTAL SYSTEMS

- **Statistical inference and learning:** Likelihood and conditional-posterior estimation; Monte Carlo marginalization over latent position and orientation; multi-observation Bayesian fusion; calibration with proper scoring rules and empirical coverage; U-Net, attention, and Transformer models.
- **Supporting RGB-D research (mentoring):** VLM-guided object proposals, depth-grounded 3D projection, cross-view association, and queryable object-centric scene memory for retrieval and localization.
- **Sequential decision-making and optimization:** Physics-informed and neuro-symbolic reinforcement learning; PPO and recurrent actor–critic policies under partial observability; bandit learning; online meta-adaptation; distributed online convex optimization.
- **Wireless digital-twin simulation and calibration:** Sionna RT scene and trajectory design; TX/RX pose and mobility simulation; geometry, material, and pose variation; measured antenna-pattern integration; stochastic RF-chain and noise modeling; Wireless InSite simulation.
- **Robotics and multimodal sensing:** *MobiFR3* system architecture and experiments; TurtleBot4/ROS 2 integration with synchronized RF, odometry, LiDAR, and RGB observations; Jackal UGV FR3 payload hardware, 3D LiDAR/RGB-D sensing, and SLAM/path-routing examples.
- **RF and software-defined radio systems:** Experiment design and supervision for a 10-GHz FR3 Pi-Radio software-defined radio (SDR) channel sounder with Vivaldi antennas, D48 pan–tilt actuation, and linear-track measurements; hands-on TI AWR1642/AWR6843 point-cloud acquisition and filtering for UAV trajectory extraction.
- **Scientific computing:** Python, PyTorch, MATLAB, R, Linux and shell scripting, LaTeX, Kubernetes; SUMO and CARLA for mobility and autonomous-driving simulation.

INDUSTRY EXPERIENCE

JD.com, Inc. *Beijing, China*
Algorithm Engineer Intern *Jan. 2021 – Feb. 2021*

- Developed CenterNet/PIoU-based rotated-object detection and supported integration and on-site validation in a manufacturing laboratory.

HONORS AND AWARDS

- Ernst Weber Fellowship, NYU Tandon School of Engineering, 2023
- Tandon Research Excellence Exhibit, *AI-Driven Interactive Safe Autonomous Driving*, 2022
- School of Engineering Fellowship, NYU Tandon School of Engineering, 2022
- China National University Student Innovation and Entrepreneurship Training Program, 2017–2019

PROFESSIONAL SERVICE

Journal reviewer: IEEE Transactions on Signal Processing; IEEE Transactions on Robotics; IEEE Transactions on Communications; IEEE Wireless Communications Letters; IEEE Transactions on Vehicular Technology; IEEE Vehicular Technology Magazine; IEEE Control Systems Letters.

Conference reviewer: IEEE International Conference on Communications (2026); Workshop on Resilience in Next Generation Networks at the IEEE International Conference on Computer Communications (2025); IEEE Global Communications Conference (2025); IEEE Conference on Decision and Control (2025); IEEE International Conference on Robotics and Automation (2024); American Control Conference (2024).

REFERENCES

Prof. Sundeep Rangan, Director, NYU WIRELESS; Professor, NYU Tandon srangan@nyu.edu
Prof. Ludovic Righetti, Professor, NYU Tandon; International Chair, ANITI ludovic.righetti@nyu.edu
Prof. JohnRoss Rizzo, MD, Associate Professor, NYU Grossman School of Medicine JohnRoss.Rizzo@nyulangone.org
Prof. Marco Mezzavilla, Associate Professor, Politecnico di Milano marco.mezzavilla@polimi.it
Prof. Tao Li, Assistant Professor, City University of Hong Kong li.tao@cityu.edu.hk